Session 2.3: Midterm Applied Analytics Formulation Part 1: Problem Structuring and Data Audit

TODAY'S PURPOSE

Why This Session Matters

- Start your midterm project
- Define one clear business question
- Confirm your data is trustworthy before analysis

Today's flow: 60 min lecture, then 120 min hands-on lab

EXPECTED OUTCOMES

By the end of today, you will be able to:

1. Frame a testable analytical hypothesis
2. Run an Exploratory Data Analysis (EDA) and data audit
3. Enforce data rules with assertion checks

Concepts for This Session

CONCEPT 1

Analytical Hypothesis Framing

Translating ambiguous managerial problems into falsifiable, testable quantitative statements expressed in executable code.

CONCEPT 2

Exploratory Data Analysis (EDA)

Systematic statistical inspection of dataset distributions, correlations, and anomalies to inform analytical design.

CONCEPT 3

Data Hygiene Audit

Automated scanning for missing records, duplicate primary keys, unexpected negative numbers, and data type inconsistencies.

CONCEPT 4

Baseline Metric Architecture

Calculating baseline benchmarks (e.g., historical contract renewal rate) against which analytical improvements are measured.

How the Concept Works in Practice

FOUNDATION 1

Translating Ambiguous Directives into Code

Senior executives often state broad challenges (e.g., "our enterprise client churn is increasing"). Data analysts must translate this into concrete quantitative questions: "Is client churn correlated with Service Level Agreement (SLA) breaches or billing discrepancies?"

FOUNDATION 2

Data Cleanliness Audit Protocols

In enterprise analytics, raw corporate data is frequently incomplete or dirty. Before building machine learning models, analysts must run programmatic audit checks to verify completeness, uniqueness, and validity.

FOUNDATION 3

Establishing Audit Baselines

Before proposing automated optimization, analysts must measure historical performance baselines. Without a baseline, proving return on investment is impossible.

FOUNDATION 4

Reproducible Script Architecture

Organizing project workspaces into clean directory structures (data/, scripts/, output/) with automated logging, ensuring any auditor can replicate findings in seconds.

Build, Test, and Diagnose in the IDE

[1] FILE EXPLORER

Organize Assets

Use the session dataset and keep scripts in a clear workspace.

[2] CODE EDITOR

Write Small Segments

Run each completed segment before expanding the program.

[3] AI AGENT CHAT

Diagnose Errors

Capture the traceback, request an explanation, and review the fix.

[4] INTEGRATED TERMINAL

Verify Execution

Confirm the final program completes with return code 0.

Script Demonstration 1: Midterm Applied Analytics Formulation Part 1: Problem Structuring and Data Audit

```
import pandas as pd

    df = pd.read_csv("data/session_02_03_dataset.csv")

    def audit_dataset_hygiene(data):
        return {
            "Total_Rows": len(data),
            "Missing_Values_Total": int(data.isnull().sum().sum()),
            "Duplicate_Accounts":
int(data["Account_ID"].duplicated().sum()),
            "Min_SLA_Compliance":
float(data["SLA_Compliance_Pct"].min()),
            "Total_Contract_Value_USD":
float(data["Contract_Value_USD"].sum())
        }

    audit_report = audit_dataset_hygiene(df)
    print("Data Hygiene Audit Report:", audit_report)
```

SYNTAX

Readable Structure

Observe the syntax and indentation that control execution.

RUNTIME

Verified Execution

Run the example in the integrated terminal and inspect the output.

BUSINESS VALUE

Applied Automation

Use the pattern to automate a repeatable business decision.

Script Demonstration 2: Midterm Applied Analytics Formulation Part 1: Problem Structuring and Data Audit

```
sla_breaches = df[df["SLA_Compliance_Pct"] < 95.0]
    print(f"Accounts Breaching SLA (<95%): {len(sla_breaches)}")
    print(sla_breaches[["Account_ID", "Enterprise_Client",
"SLA_Compliance_Pct", "Client_Retention_Risk"]])
```

SYNTAX

Readable Structure

Observe the syntax and indentation that control execution.

RUNTIME

Verified Execution

Run the example in the integrated terminal and inspect the output.

BUSINESS VALUE

Applied Automation

Use the pattern to automate a repeatable business decision.

Script Demonstration 3: Midterm Applied Analytics Formulation Part 1: Problem Structuring and Data Audit

```
risk_summary = df.groupby("Client_Retention_Risk").agg(
        Total_Contract_Value_USD=("Contract_Value_USD", "sum"),
        Account_Count=("Account_ID", "count"),
        Mean_Billing_Discrepancies=("Billing_Discrepancy_Count",
"mean")
    )
    print("Client Retention Financial Exposure Summary:\\n",
risk_summary)
```

SYNTAX

Readable Structure

Observe the syntax and indentation that control execution.

RUNTIME

Verified Execution

Run the example in the integrated terminal and inspect the output.

BUSINESS VALUE

Applied Automation

Use the pattern to automate a repeatable business decision.

Exercise 1 and Exercise 2

EXERCISE 1: GUIDED

Foundational Practice

Follow the lab protocol, run the script, and verify its output with the session dataset.

EXERCISE 2: DEBUGGING

Applied Challenge

Extend the program, diagnose an intentional error, apply the verified correction, and confirm return code 0.